FH Innovations Inc. (FHI)

Enterprise Architecture Proposal

An Enterprise Architecture Vision, Framework, and Blueprint

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Framework Reference: The Open Group Architecture Framework (TOGAF), 10th Edition

1. Executive Summary

FH Innovations Inc. (FHI) is an automotive services MSME with approximately 27 employees, specializing in decarbonizing, aircon treatment, engine treatment, and nitrogen tire inflation services delivered on-site at partner dealership service bays such as Toyota Balintawak and Toyota Manila Bay. FHI currently operates entirely on manual record-keeping — index cards and fragmented spreadsheets — with no dedicated IT infrastructure or IT personnel.

The central problem the architecture addresses is operational friction and working-capital compression caused by four gaps:

1. No reconciliation between items purchased and items delivered to dealer consignments
2. Recurring stockouts driven by the geographic distance of FHI's suppliers
3. Unsystematic Accounts Receivable (AR) aging and long-outstanding balances across multiple disconnected bank accounts
4. Manual, unverified pre-numbered Job Orders that create human error and untraceable cancellations.

The proposed To-Be architecture replaces these manual touchpoints with a lightweight, cloud-based Inventory & Invoicing / ERP solution built around five centralized data entities and three enforceable architecture principles.

2. Preliminary Phase — Framework and Principles

Boundary and Scope Statement

- Organizational Units Impacted: FHI corporate administration, finance, and field service operations, including interaction touchpoints with external partner dealership service bays (e.g., Toyota Balintawak, Toyota Manila Bay).
- Geographic Locations: FHI headquarters and operational touchpoints across Metro Manila automotive service centers.
- Functional Scope: End-to-end operational visibility across Procurement, Consignment Inventory Tracking, Job Order Execution, Billing, and Accounts Receivable (AR) Management.
- Exclusions: Direct structural or technical changes to external partner dealership networks (e.g., Toyota's internal Dealer Management Systems). The architecture is scoped strictly to FHI's own interface touchpoints, delivery handoffs, and consignment ledgers.

Framework Interface Assessment

FHI has no formal IT department or software development lifecycle today. The EA framework therefore interfaces directly with existing business-administration rhythms rather than software:

- ADM Cycle Synchronization: ADM iterations are aligned with FHI's monthly financial closing and billing reconciliation cycle managed by Ms. Lisa (Administrator / Business Partner), so architecture milestones track the business's natural rhythm.
- Operational Integration: Transition milestones are phased to avoid disrupting daily decarbonizing and vehicle-treatment services at partner dealership bays. Change is introduced incrementally — starting with administrative data logging before expanding to real-time mobile technician tracking.

Tailored Enterprise Metamodel

FHI's lightweight operational structure calls for a streamlined subset of the standard TOGAF Foundation Metamodel, stripping out enterprise-grade constructs that do not apply to a flat, 27-person MSME:

Retained entities — Business Actor, Business Role, Business Service, Business Process, and Business Event; Data Object, Application Component, and Application Service; Physical Technology Component and System Software.

Deleted entities — Logical Technology Component, Technology Service, external enterprise-grade Contract/SLA constructs, Location (beyond FHI's direct physical service bays), and Organization Unit Hierarchy (beyond FHI's flat structure).

Tailored Content Framework

The following artifacts from the TOGAF Extended Content Framework will be produced during the corresponding ADM phases. They are referenced by name here and produced in full in Sections 4–6 of this document.

Phase B — Business Architecture artifacts

- Business Capability Map — core services: Decarbonizing, Aircon/Engine Treatment, Nitrogen Inflation.
- Process Flow Diagram (As-Is vs. To-Be) — Job Order issuance, dealer routing, and consignment reconciliation.
- Actor-to-Role Matrix — responsibilities across FHI administrative staff, service technicians, and dealer service advisors.

Phase C — Information Systems Architecture artifacts

- Data Entity / Data Flow Diagram — Purchase Order > Delivery Receipt > Consignment Usage > Billing Statement > AR Ledger.

- Application Communication Diagram — interaction between inventory logs, invoicing software, and multi-bank tracking.
- Information Use-Case Matrix — mapping data objects (Job Orders, POs, AR aging brackets) to business roles.

Phase D — Technology Architecture artifacts

- Technology Infrastructure Catalog — baseline administrative computers, field tablets, cloud backup systems.
- Network & Data Topology Diagram — secure cloud synchronization paths from field devices to administrative dashboards.

Principles Catalog

Principle 1 — Digital-First Over Manual Tracking

Statement: All business transactions, inventory movements, job orders, and billing metrics must be recorded digitally at the point of origin, replacing manual index cards and fragmented spreadsheets.

Rationale: Manual tracking introduces high rates of human error, misplaced records, and delayed billing reconciliations that directly compress FHI's working capital and create opaque operational visibility.

Implications: FHI staff must be equipped with mobile or desktop input tools; basic digital data-entry standards and training must be institutionalized across all 27 employees.

Principle 2 — Three-Way Consignment Reconciliation

Statement: Every item delivered to a partner dealership must maintain verifiable traceability across Purchase Orders, Delivery Receipts, and actual Jobs Done before billing generation.

Rationale: Current discrepancies between items bought and items delivered lead to uncollected revenue and inventory shrinkage across distributed dealer locations (e.g., Toyota Balintawak, Manila Bay).

Implications: Strict pre-numbered Job Order and delivery-receipt verification protocols, managed jointly by field staff and administration, so no service is rendered without digital logging.

Principle 3 — Automated Accounts Receivable Aging

Statement: Financial workflows must systematically track invoice issuance, collection windows, and aging brackets to trigger structured follow-up mechanisms automatically.

Rationale: Long outstanding payments severely impact FHI's cash flow, exacerbated by maintaining accounts across multiple disconnected banks and relying on reactive follow-up letters.

Implications: Adoption of centralized billing dashboards with automated aging alerts, scheduled collection windows, and streamlined bank-deposit reconciliation logs.

3. Phase A — Architecture Vision

Business Goals and Objectives

- Process Automation: Replace manual index cards and fragmented spreadsheets with centralized digital workflows for Purchase Orders (POs), Job Orders (JOs), and billing.
- Inventory & Consignment Reconciliation: Establish robust tracking between items purchased, items delivered to partner dealerships, and services rendered.
- Working Capital Optimization: Implement systematic AR aging tracking and automated follow-up workflows to reduce outstanding balances and streamline collection days.
- Supply Chain Resilience: Mitigate geographic bottlenecks (Balintawak supplier distance) through automated inventory reorder alerts and supplier communication channels.

FH Innovations Inc. (FHI) exists to provide specialized automotive services through partner dealership service bays. It creates value by delivering services such as decarbonizing, aircon and engine treatment, and nitrogen tire inflation. It serves customers through partner dealerships and generates revenue from the automotive services it performs, which are documented through Job Orders and processed for billing and collection.

The company's strategy is to provide specialized automotive services through partner dealerships while maintaining inventory availability, processing completed services for billing, and collecting payments to sustain operations

The business strives to not only make profit but also to provide the best service. Despite the lack of IT infrastructure for record-keeping, they still deliver value to both their customers and stakeholders.

Stakeholder Map and RACI Matrix

The table below consolidates each stakeholder's architectural interest with their responsibility level (RACI) across the ADM phases covered in this document.

Stakeholder	Architectural Interest / Concerns	Phase A	Phase B	Phase C	Phase D	Phase E/F
Harry Ho— President	Strategic visibility, overall financial health, risk mitigation, and scaling operations.	A	A	C	C	A
Ms. Lisa — Administrator / Business Partner	Consignment billing accuracy, collection tracking, dealer handoffs, reducing manual reconciliation overhead.	R	R	R	C	R

Stakeholder	Architectural Interest / Concerns	Phase A	Phase B	Phase C	Phase D	Phase E/F
FHI Service Staff	Real-time job order status updates, inventory availability, decarbonizing execution logs, eliminating stockouts.	C	R	R	R	C
Partner Dealers (Toyota Balintawak, Toyota Manila Bay, etc.)	Timely repair request processing, accurate pre-numbered job order generation, clear billing statements.	I	C	I	I	I
Lead EA / Consultant	Architecture framework execution, process optimization, system deployment.	R	R	R	R	R

Phase A = Vision, Phase B = Business, Phase C = Information Systems, Phase D = Technology, Phase E/F = Migration Planning / Governance.

R = Responsible A = Accountable C = Consulted I = Informed

Request for Architecture Work

Sponsors: Harry Ho (President, FHI) and Ms. Lisa (Administrator, FHI).

Triggering Event: Operational friction — stockouts caused by Balintawak supplier distance, manual reconciliation bottlenecks, and uncollected long-outstanding accounts across 27 employees and multiple dealership partners.

Business Goals: Eliminate manual index-card and spreadsheet record-keeping vulnerabilities; streamline the consignment billing and collection pipeline to improve cash flow; establish robust inventory reconciliation between procurement and multi-dealer delivery points.

Major Organizational Constraints: Small team scale (~27 employees) with no dedicated IT personnel, limited initial capital expenditure capacity requiring lightweight, cost-effective tooling; geographic dispersion across partner dealership service bays requiring low-friction mobile or cloud-accessible processes.

Project Scope & Boundaries: Excludes structural changes to partner car dealerships; targets strictly internal FHI administrative processes, supplier ordering loops, and field inventory reconciliation touchpoints.

Business Drivers:

- Demand for automotive treatment services
- Partner dealership relationships
- Reliable service delivery
- Availability of supplies and inventory

- Revenue collection and cash flow
- Ability to maintain and scale operations

4. Phase B — Business Architecture

As-Is vs. To-Be Process Comparison

Domain	As-Is Architecture (Manual)	To-Be Architecture (Digitized)
Procurement & Inventory	Manual phone calls to suppliers, reactive purchasing, no tracking of items bought vs. delivered.	Automated stock monitoring; digital PO generation; automated reorder triggers for decarbonizing solutions.
Service Execution	Manual repair requests routed across dealers, paper job orders prone to completeness gaps and cancellations.	Digital pre-numbered Job Order creation tied directly to vehicle details (Make, Model, Plate No.) and synchronized in real time.
Billing & Consignment	Manual data collection via index cards/spreadsheets, delayed billing cycles tied to vehicle release status.	Automated billing triggers upon verified job completion; digital consignment ledger tracking units sold vs. remaining.
Collections & Accounts	Manual tracking of outstanding accounts across multiple bank accounts, reactive follow-up letters.	Integrated AR aging reports, automated follow-up schedules, and centralized bank reconciliation.

Core Value Stream

The value stream traces a single service engagement from supplier procurement through to cash collection:

- Procurement — Supplier PO fulfillment for chemical and consumable treatments.
- Execution — Service request issuance by dealer Service Advisors > FHI staff execution of decarbonizing and related services.
- Verification — Pre-numbered Job Order validation and administrative collection.
- Billing & Collection — Consignment statement generation, AR aging management, and multi-bank payment reconciliation.

Business Capability Map

FHI's business capabilities cluster around three revenue-generating service lines, each underpinned by the same shared support capabilities:

- Core service capabilities: Decarbonizing Treatment, Aircon/Engine Treatment, Nitrogen Tire Inflation.
- Shared support capabilities: Inventory & Consignment Management, Job Order Administration, Billing & Collections, Supplier & Dealer Relationship Management.

Actor-to-Role Matrix

Actor	Role	Key Responsibility in the To-Be Process
FHI Administrator (Ms. Lisa)	Business Partner / Billing Owner	Validates Job Orders, issues consignment statements, manages AR aging and bank reconciliation.
FHI Service Technician	Field Operations	Executes services on-site, logs digital Job Orders, flags stock needs in real time.
Dealer Service Advisor	External Requestor	Issues repair/service requests and confirms vehicle details for Job Order creation.
FHI President (Harry Ho)	Strategic Owner	Approves architecture direction, reviews financial health and risk indicators.

5. Phase C — Information Systems Architecture

Data Architecture

To resolve current data silos, FHI must centralize master data across five core entities:

Inventory Master — tracks stock levels of decarbonizing solutions, aircon/engine treatments, and nitrogen supplies across storage and dealer consignments.

Dealer & Partner Master — maintains profiles, contact details, and routing data for partner service centers (Toyota Balintawak, Toyota Manila Bay).

Job Order (JO) Ledger — captures pre-numbered transaction metadata: Vehicle Make, Model, Plate Number, Service Type, Status (Completed, Pending Further Repairs, Cancelled), and Assigned Staff.

Consignment & Billing Database — manages items delivered versus items sold, preventing discrepancies and serving as the direct input for billing statements.

Accounts Receivable (AR) Ledger — tracks invoice issue dates, payment terms, collection schedules, aging brackets (0–30, 31–60, 61+ days), and multi-bank deposit logs.

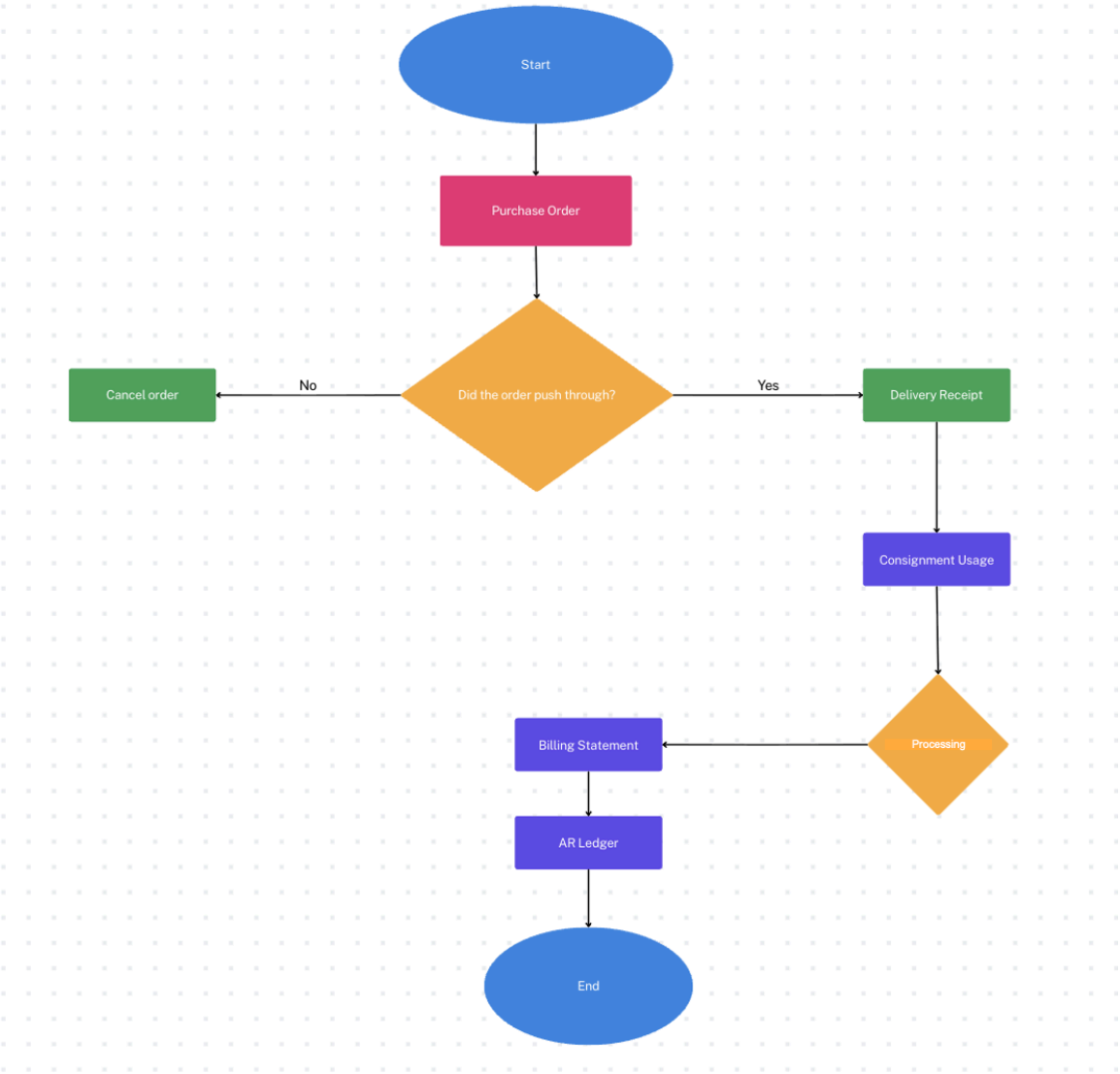
Application Architecture

Given FHI's status as an MSME with 27 employees and no existing IT architecture, a lightweight, cloud-based or local-server-hosted ERP or Inventory & Invoicing SaaS is recommended, comprising three modules:

- Inventory Management Module — real-time stock visibility and delivery-receipt tracking.
- Service & Job Order Module — mobile/tablet-friendly digital job logging for technicians and admin staff.
- Billing & AR Module — automated statement generation and aging reports with collection-calendar alerts.

Data Entity / Data Flow

Data flows in a single directional chain that mirrors the Three-Way Consignment Reconciliation principle:



Application Communication Diagram (Narrative)

The Inventory Management Module publishes stock and delivery events consumed by the Service & Job Order Module, which in turn confirms completed services to the Billing & AR Module. The Billing & AR Module reconciles against multi-bank deposit feeds to close the loop from job completion to cash collection.

Information Use-Case Matrix

Data Object	Primary Business Role(s)	Use Case
Purchase Orders (PO)	Administrator, President	Trigger and track supplier procurement.
Job Orders (JO)	Service Technician, Administrator	Log and verify service execution per vehicle.
Consignment Ledger	Administrator	Reconcile items delivered vs. items sold per dealer.
AR Aging Buckets	Administrator, President	Drive collection follow-up and cash-flow visibility.

6. Phase D — Technology Architecture

Infrastructure Transition Plan

Hardware Provisioning — procure baseline office computing hardware (desktops/laptops for Admin) and mobile tablets/ruggedized devices for field service staff to log jobs on-site.

Connectivity & Cloud Access — implement reliable high-speed internet connectivity at the main office to support cloud-based application synchronization, reducing reliance on manual physical handoffs.

Data Security & Backup — establish daily automated backups of financial, billing, and operational records to replace physical index-card reliance and prevent data loss.

Technology Infrastructure Catalog

Component	Type	Purpose
Admin Desktop/Laptop	Physical Technology Component	Office-based data entry, billing, and AR management (Ms. Lisa).
Field Tablet / Ruggedized Device	Physical Technology Component	On-site Job Order logging by service technicians.
Cloud Inventory & Invoicing SaaS	System Software	Central application hosting Inventory, Job Order, and Billing/AR modules.
Automated Backup Service	System Software	Daily backup of financial and operational records.

Network & Data Topology

Field tablets synchronize Job Order and inventory data to the cloud application over the office's high-speed internet connection whenever devices are in range of connectivity; the central administrative dashboard consumes this synchronized data to drive billing and AR workflows, with automated daily backups running independently of the operational sync path.

7. Gap Analysis and Risk Mitigation

Identified Operational Gap	Architectural Impact	Proposed Remediation Strategy
No reconciliation between items bought and delivered	Inventory leakage, financial discrepancies, unaccounted stock.	Implement a digital 3-way match protocol: Purchase Order vs. Delivery Receipt vs. Consignment Usage Report.
Distance problem with Balintawak supplier causing stockouts	Operational downtime, staff forced to scramble for stock.	Implement safety-stock thresholds with automated lead-time buffers and scheduled bulk ordering.
Unsystematic AR aging and long outstanding payments	Cash-flow compression, delayed follow-ups, manual statement creation.	Introduce automated AR aging dashboards with rule-based triggers for follow-up letter generation and structured collection days.
Manual tracking via index cards and unverified pre-numbered forms	Human error, missing job orders, inability to track cancellations.	Enforce mandatory digital validation of pre-numbered delivery receipts and Job Orders prior to closing out service tickets.

8. Conclusion and Next Steps

This unified architecture takes FHI from a manual, high-friction operating model to a digitized, traceable one, using a metamodel and content framework deliberately tailored to a flat, 27-person MSME rather than a full enterprise-grade deployment. The three architecture principles — Digital-First Over Manual Tracking, Three-Way Consignment Reconciliation, and Automated AR Aging — give a small, enforceable set of rules that resolve the four operational gaps identified without requiring a dedicated IT department.

Recommended next steps are to secure sponsor sign-off on the Request for Architecture Work, select a lightweight cloud-based Inventory & Invoicing SaaS consistent with the Application Architecture, and pilot the digital Job Order workflow at a single dealer partner (e.g., Toyota Balintawak) before rolling out across all partner dealerships — consistent with the phased, non-disruptive Operational Integration approach defined in the Framework Interface Assessment.